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Evaluation of Steam Explosion Pretreatment and Enzymatic Hydrolysis Conditions for Agave Bagasse in Biomethane Production

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Abstract

The production of biofuels from lignocellulosic biomass includes a pretreatment step to alter the biomass structure and facilitate the enzymatic degradation of the polymers to obtain assimilable compounds. In this study, agave bagasse (AB) was used as a feedstock for obtaining methane, for which AB was pretreated with steam explosion and enzymatically hydrolyzed. The pretreatment conditions corresponded to severity factors (SFs) within a range from 1.65 to 2.89, while enzymatic hydrolysis was performed with enzyme loads of Cellic CTec2 within a range from 0.12 to 3.6 $mg_{protein}\ g^{-1}{}_{AB}$. The best global yields (including pretreatment and enzymatic hydrolysis) of total carbohydrates (TCs), glucose (GLU), xylose (XYL), and chemical oxygen demand (COD) were 0.7 g TC $g^{-1}{}_{AB}$, 0.12 g GLU $g^{-1}{}_{AB}$, 0.03 g XYL $g^{-1}{}_{AB}$, and 0.20 g O_2 $g^{-1}{}_{AB}$ obtained using 2.4 $mg_{protein}\ g^{-1}{}_{AB}$ of Cellic CTec2 with agave bagasse pretreated with an SF of 2.41. The contribution of pretreatment to the global TC yield ranged from 13 to 34% for the different systems evaluated. The biochemical potential of methane (BMP) of hydrolysates (pretreatment at SF 2.41 and 2.4 $mg_{protein}\ g^{-1}{}_{AB}$ of Cellic CTec2) was 0.284 ± 0.02 in NL CH_4 g^{-1} COD with a COD removal of 78.4 ± 1.3. This BMP value was 40% higher than the BMP obtained in the system without enzymatic hydrolysis, indicating the impact of this step on conversion to biomethane. The results at the BMP level indicated the potential of this residue for biofuel production.

Keywords Biochemical methane potential (BMP) · Pretreatment · Lignocellulosic biomass · Sugar recovery · Biogas · Severity factor

Introduction

Lignocellulosic agricultural wastes, which are mostly discarded or disposed of, resulting in a variety of environmental problems, can be used as feedstock for second-generation biofuel production. The growing interest in biofuel production is related to a concern about the depletion of oil reserves and a possible crisis due to fuel shortages and, from an environmental point of view, due to the potential reduction of greenhouse gas emissions [1].

In Mexico, the projections by 2030 for the development of second-generation biofuels include at least the use of 4% in transportation and 35% in electricity [2]. Agave bagasse (AB) is a lignocellulosic residue from tequila production with potential as feedstock since it is one of the most important mexican industries, generating approximately 540,000 tons of AB per year that are disposed of in fields or burned at tequila factories [3].

Lignocellulosic biomass has a recalcitrant structure due to its composition of cellulose, hemicellulose, and lignin; therefore, pretreatment and enzymatic hydrolysis are required to obtain carbohydrates that can be converted into biofuels. Pretreatment is a critical step since the structure of lignocellulosic matter is altered to facilitate the access of enzymes to chains of polymers. Physicochemical, chemical, and biological pretreatments can be applied to lignocellulosic biomass; among these pretreatments, one of the most studied in the laboratory, demonstration, and commercial plants is steam explosion [4]. In this hydrothermal pretreatment, the biomass is processed with saturated steam at temperatures between 160 and 280 °C for times ranging from seconds to minutes, followed by sudden depressurization [5, 6]. The main effects of the steam explosion process on biomass are as follows: (a)

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reduction of the degree of polymerization due to acidic degradation and thermal degradation; (b) mechanical breakage of fibers to a certain extent due to the sudden decompression of the system, the consequent deconstruction of the amorphous regions, and the partial destruction of the crystalline regions; and (c) destruction of the hydrogen bonds and structural rearrangement due to temperature and pressure conditions [7, 8]. In addition to the above, this pretreatment uses almost no chemical products since it requires only water to solubilize and depolymerize the hemicellulose in the liquid phase (hydrolysates). In this way, steam explosion can be considered an environmentally friendly process and was selected for this study [9]. Other advantages of steam explosion include a significantly lower environmental impact and a lower capital investment compared with other leading pretreatments. However, the main disadvantage is the energy requirement to achieve the operational conditions. However, it has recently been shown that steam explosion pretreatment of AB for the production of methane is energetically feasible [3, 8]. After pretreatment, enzyme complexes, mainly cellulases and hemicellulases, act on the β-1,4-glycosidic bonds that join the cellulose and hemicellulose molecules to release the monomers, allowing further processing [10].

The potential of AB for the production of gaseous biofuels has been explored by Valdez-Vazquez et al. [11], who reported a comparison of biological, enzymatic, chemical, and hydrothermal pretreatments. Similarly, Galindo-Hernández et al. [12] pretreated AB with alkaline hydrogen peroxide and performed enzymatic saccharification to evaluate hydrogen and methane production. García-Amador et al. [13] evaluated the feasibility of bioelectrohydrogen production from steam explosion–pretreated AB. Other studies such as ethanol performing autohydrolysis at severity factors (SF) ranging from 2.35 to 4.12, as well as enzymatic hydrolysis (EH) and fermentation with *S. cerevisiae* [14], have been performed for the production of liquid biofuels.

In this study, AB was pretreated by steam explosion with temperatures below the temperatures commonly reported in the literature, which resulted in less severe conditions and lower energy requirements. In this context, the objective of this work was to evaluate the yields of total carbohydrates (TCs), glucose (GLU), and xylose (XYL) obtained by enzymatic hydrolysis of steam explosion–pretreated agave bagasse under different operating conditions and enzymatic cocktails for their later conversion to biomethane.

Methodology

Biomass

Bagasse of *Agave tequilana* Weber, blue variety, obtained from a tequila factory from Amatitán, state of Jalisco, was donated by the Institute for Scientific and Technological Research of San Luis Potosi (IPICYT). The bagasse was sun-dried, sieved (< 1.7 mm), and stored at room temperature [15]. The composition of AB was 46.0 ± 1.0% cellulose, 23.1 ± 1.0% hemicellulose, 15.3 ± 1.0% lignin, 6.8 ± 1.4% total extractives, 4.3 ± 1.2% moisture content, and 2.9 ± 1.2% ash [3].

Pretreatment

Unground dried agave bagasse was pretreated by steam explosion at a solid:liquid ratio of 55:2000 (wt/wt) and temperatures of 116, 142, and 154 °C for 15 and 20 min. These operational conditions were selected based on our previous study, where the energy balance at different operational conditions was evaluated [3].

The SF was used for the comparison of pretreatments since it includes the operational parameters of temperature and time of pretreatment and is consequently related to the energy requirements. The SF was calculated for each pretreatment according to Eq. 1, which is based on first-order kinetics and obeys Arrhenius's law [16]. The conditions of the pretreatments employed are shown in Table 1.

$$\mathrm{SF} = \mathrm{Log}\left(t^{*}e^{\frac{(T-100)}{14.75}}\right) \quad (1)$$

where T is the temperature in °C, and t is the time in minutes.

Two fractions were obtained from the pretreatment, one solid fraction consisting of the pretreated (humid) bagasse and one liquid fraction, the hydrolysates. The solid fraction was dried at 38 °C for preservation, whereas the hydrolysates were refrigerated at 4 °C until used for EH (Table 2). As described below, the solid fraction was characterized after acid hydrolysis [17] and analyzed, as were the hydrolysates.

Table 1 Experimental conditions and severity factor values of the pretreatments

Temperature (°C)	Pressure (MPa)	Time (min)	SF
116	0.28	15	1.65
142	0.47	15	2.41
142	0.47	20	2.54
154	0.67	15	2.77
154	0.67	20	2.89

SF, severity factor

Solid: liquid ratio (wt/wt) in all cases 55:2000

Table 2 Experimental conditions for enzymatic hydrolysis

SF	Enzyme	Concentration ($mg_{protein}\ g^{-1}{}_{AB}$)	Aqueous phase
Control	Cellic CTec2	0.12-2.40	0.1 M Acetate Buffer, pH 5
2.54 2.77	Cellic CTec2	0.12	0.1 M Acetate Buffer, pH 5
2.89			
2.41 2.54	Cellic CTec2	0.12	Hydrolysates, pH 5
2.41	Cellic CTec2	1.20	
1.65 2.41[a]	Cellic CTec2	2.40	
2.54			
2.77			
2.89			
2.41	Cellic CTec2	3.60	
2.41[a]	Celluclast 1.5 L-Viscozyme L	3.7–19.4	50 mM Citrate Buffer, pH 3.5

[a] Selected for BMP tests; *SF*, severity factor

Enzymatic Hydrolysis

The EH of the solid fraction obtained with the different pretreatments studied was evaluated using buffer and different concentrations of the Cellic CTec2 cocktail (Novozymes), as shown in Table 2. Simultaneous EH of the solid and liquid fractions resulting from the different pretreatments studied was also performed as described in Table 2. The aqueous phase obtained after EH is referred to as enzymatic hydrolysates, and its BMP was evaluated for selected experiments (described later). In the first set of experiments, an enzymatic concentration of 0.12 $mg_{protein}\ g^{-1}{}_{AB}$ with different SF levels was evaluated; in another set of experiments, different concentrations of the Cellic CTec2 cocktail were tested with materials pretreated with SF 2.41. Finally, the concentration of 2.4 $mg_{protein}\ g^{-1}{}_{AB}$ of Cellic CTec2 with different SF levels was tested. Temperature, pH, stirring speed, and time were set according to the manufacturers' specifications at 50 °C, initial pH 5, 120 rpm, and 34 h, using a ratio of 25 g_{AB} per liter of buffer or hydrolysate. When needed, the initial pH of 5 was adjusted using 0.1 M NaOH; there was no significant change in pH during EH, and it was therefore not controlled.

A comparison with the enzymatic mix consisting of Celluclast 1.5 L-Viscozyme L mixture (Novozymes) donated by Dr. Felipe Alatriste-Mondragón from IPICYT was used. This mixture has previously been optimized for AB hydrolysis using the same AB batch as in this study [15]. Therefore, the enzymatic concentration and the operational conditions were fixed at the optimized values: 3.7 $mg_{protein}\ g^{-1}{}_{AB}$ of Celluclast 1.5 L, 19.4 Viscozyme L, 40 °C, and 120 rpm for 12 h, with a ratio of 64.84 g_{AB} per liter of the buffer (Table 2).

Control assays were carried out with AB without pretreatment under the conditions mentioned previously for each enzymatic cocktail. Negative control assays were carried out without adding AB under the same conditions. All experiments were performed in triplicate.

Biochemical Methane Potential Tests

Based on the results obtained, the enzymatic hydrolysates obtained with Cellic CTec2 at 2.4 $mg_{protein}\ g^{-1}{}_{AB}$, those obtained with Celluclast 1.5 L-Viscozyme L, and the control without EH were selected to be subjected to a BMP test, and the three were pretreated with SF 2.41. These analyses were carried out at the Center for Exact Sciences and Engineering (CUCEI) of the University of Guadalajara using the Automatic Methane Potential Test System (AMPTS II, Bioprocess Control, Sweden). Anaerobic sludge from a UASB reactor employed to treat tequila vinasses was used as inoculum for anaerobic digestion under the following conditions: 37 °C, 150 rpm, initial pH of 7.5, 10 g volatile solids total per liter of sludge as inoculum, and the initial COD values were fixed to 5 g COD L^{-1} [18].

Analytical Methods

The protein concentration of the enzymatic cocktails was determined by the Lowry assay according to the Instruction Manual, DC Protein Assay and quantified by spectrophotometry at 750 nm (Biotraza, model 752) [19].

Periodic aqueous samples of enzymatic hydrolysates were taken from each experiment. The analyses performed were as follows: (a) TC by the phenol-sulfuric acid method and quantified by spectrophotometry at 485 nm (Biotraza, model 752) [20], (b) GLU and XYL concentrations using a Biochemistry Analyzer YSI-2700, and (c) COD according to the HACH

Method 8000 using a Hach DRB200 reactor for COD digestion and quantified spectrophotometrically at 600 nm (Biotraza, model 752) [21]. Acetic acid was quantified by high-performance liquid chromatography (HPLC, Varian ProStar 210) with a UV–vis detector (Varian ProStar PS 325, 278 nm) using an AMinex HPX-87H column [3].

Yield Calculations and Data Analysis

The EH yields of TC, GLU, and XYL expressed as g $g^{-1}{}_{AB}$ were calculated by subtracting the initial concentration of carbohydrates from the final carbohydrate concentrations and considering the volume of the hydrolysates and the mass of AB on a dry basis. The global yield included sugar recovery from both the pretreatment and EH, in all cases referring to the initial content in the AB.

The experimental EH and BMP data were fitted by the Gompertz model using OriginPro 8 software. This logistic model has the parameters A, t_r, and k related by Eq. 2.

$$Sc = Ae^{-e(-k(t-t_r))} \tag{2}$$

where Sc is the TC concentration (g L^{-1}) or methane concentration (NmL); k is the production rate (h^{-1}); A is a parameter related to final concentration values; t is time (h); and t_r is the time at the inflection point of the curve (h). The maximum production rate (V_{max}) can be estimated by derivation of Eq. 2, resulting in Eq. 3:

$$V_{max} = 0.386\,A\,k \tag{3}$$

One-way analysis of variance (ANOVA) was performed, and statistical significance was determined at a level of $p <$ 0.05 using IBM SPSS software.

Results and Discussion

Total Carbohydrates After Pretreatment

The characterization of the pretreated bagasse and the hydrolysates obtained after steam explosion is shown in Table 3. In all cases, higher TC yields were found in the residual solid fraction (0.59 to 0.76 g $g^{-1}{}_{AB}$) than in the hydrolysates (0.04 to 0.08 g $g^{-1}{}_{AB}$) for the different pretreatments tested. The glucose content in the solid fraction after pretreatment ranged from 0.54 to 0.75 g GLU g^{-1} TC (Table 1S in Electronic Supplementary Material), indicating that they were the main carbohydrates present in that fraction. These TC contents were affected by the severity of the pretreatment; the highest value was found at the lowest SF, while the lowest value was observed under the most severe conditions (Table 3). The sums of the solid fraction and the hydrolysates were not significantly different, except for more severe conditions (SF 2.89), which could be attributable to the degradation of sugars by acid hydrolysis or temperature [6]. However, 0.04 to 0.05 g g^{-1} TC, GLU, and XYL were in the hydrolysates. Therefore, the main effect of the pretreatment was found in the hemicellulose fraction, whereas the cellulose fraction mainly remained in the solid fraction, requiring further hydrolysis via enzymatic reactions. This approach was also supported by the presence of acetic acid (0.45–0.55 g L^{-1}) in the liquid fraction of all pretreatment conditions (Table 1S in the Electronic Supplementary Material). The formation of this compound during pretreatment has been related to xylan deacetylation after depolymerization, and as a result of these reactions, accessibility to carbohydrates in the solid residual fraction is favored [3, 22].

Effect of SF on Enzymatic Hydrolysis

The highest TC yield obtained from the EH (0.12 $mg_{protein}$ $g^{-1}{}_{AB}$ Cellic CTec2) of the solid fraction pretreated at SF 2.89 was 13 and 27% higher than the TC yield of the control and SF 2.54, respectively (Fig. 1). The GLU yield obtained at SF 2.54 was 15 to 188% higher than the GLU yields of the other SFs. However, the XYL yield remained the same for the control as well as SF 2.54 and 2.89 pretreatments, in agreement with the characterization of the solid fraction and hydrolysates after pretreatment (discussed in the previous section), where we observed that cellulose remained mainly in the solid fraction, while hemicellulose was released as disaccharides and oligosaccharides in the hydrolysates. The increment in SF from 2.54 to 2.89 (due to an increment of temperature from 142 to 154 °C) promoted a higher TC yield, but the GLU yield was reduced at SF 2.89 compared with 2.54, indicating the possible recovery of other carbohydrates and the modification of the AB fibers due to the higher temperature resulting in the degradation of structural sugars. Furthermore, when using both fractions (liquid and solid) resulting from the pretreatment at SF 2.54, the TC and XYL yields increased by 40 and 155%, respectively, compared to the experiments where only the solid fraction was enzymatically hydrolysate, while no

Table 3 TC contents in the pretreated bagasse and hydrolysates after pretreatment

	Solid fraction	Hydrolysates	
SF	TC (g $g^{-1}{}_{AB}$)		pH
1.65	0.76 ± 0.03	0.040 ± 0.001	5.1 ± 0.1
2.41	0.66 ± 4 × 10^{-3}	0.055 ± 2 × 10^{-4}	4.0 ± 0.1
2.54	0.74 ± 0.02	0.057 ± 0.006	4.1 ± 0.1
2.77	0.61 ± 0.04	0.079 ± 0.007	4.7 ± 0.1
2.89	0.59 ± 0.02	0.073 ± 0.009	4.8 ± 0.1

SF, severity factor, *TC*, total carbohydrates

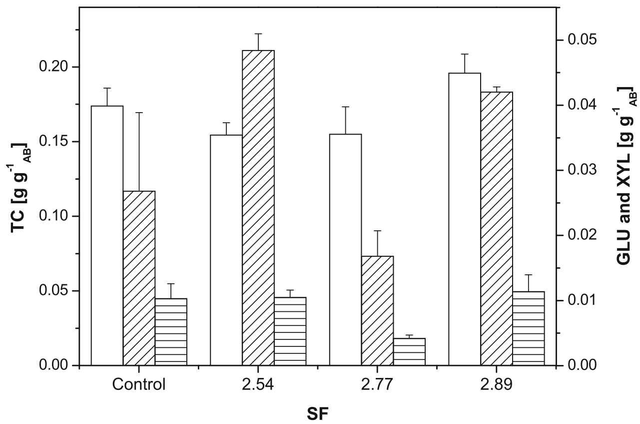


Fig. 1 Comparison of sugar yields after enzymatic hydrolysis with 0.12 $mg_{protein}$ $g^{-1}{}_{AB}$ with different SF values. (□) TC; (▨) GLU; (▤) XYL

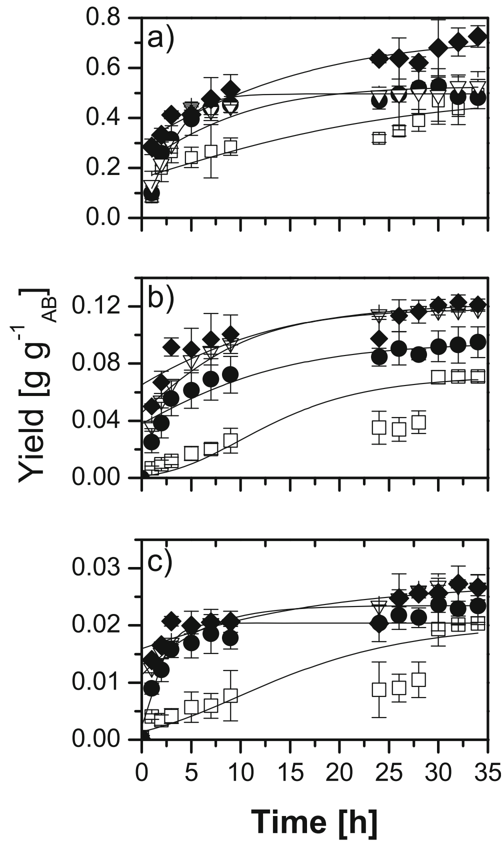


Fig. 2 Yields of (**a**) TC, (**b**) GLU, and (**c**) XYL of the enzymatic hydrolysis of AB pretreated at SF 2.41 at Cellic CTec2 concentrations in $mg_{protein}$ $g^{-1}{}_{AB}$ of 0.12 (□), 1.2 (●), 2.4 (▽), and 3.6 (◆). Gompertz model fit (continuous line)

significant difference was observed for GLU yield (data not shown).

The TC yields obtained (Fig. 1) are comparable with the TC yields reported previously (~ 0.15 ± 0.02 g TC $g^{-1}{}_{AB}$) when 0.7 $mg_{protein}$ $g^{-1}{}_{AB}$ of Celluclast 1.5 L was used [23]. However, the values of 0.19 ± 0.02 g TC $g^{-1}{}_{AB}$ obtained for SF 2.89 are comparable with the values obtained when using alkaline peroxide hydrogen as pretreatment and EH (20 $mg_{protein}$ $g^{-1}{}_{AB}$ of Celluclast 1.5 L) [12]. However, the GLU yields were six times lower than the GLU yields reported in a study using ionic liquid pretreatment and EH with Cellic CTec2 (70 $mg_{protein}$ $g^{-1}{}_{AB}$) [24]. Therefore, we applied increased enzyme concentrations, as discussed in the following section.

Effect of Enzyme Concentration

Figures 2a, b, and c show the evolution of TC, GLU, and XYL concentrations obtained during the EH of pretreated AB at SF 2.41 and Cellic CTec2 concentrations tested. Increasing the enzymatic concentration from 0.12 and 1.2 $mg_{protein}$ $g^{-1}{}_{AB}$ (10X) did not result in significantly different TC, GLU, or XYL yields. However, the increase in enzymatic concentration by 20X and 30X increased the final TC concentration by 8 and 48%, respectively. A similar behavior was observed for the GLU and XYL concentrations with the 20X enzymatic concentration, where significant differences of 24 and 16% were found. The GLU and XYL yields obtained at the highest enzymatic concentration tested were not significantly different from those obtained with 20X (GLU and XYL yields of 0.12 and 0.03 g $g^{-1}{}_{AB}$, respectively).

The fit of these experimental data to the Gompertz model (R^2 0.904 to 0.990) is shown in Fig. 3. In general, the increase in the enzymatic concentration resulted in an increase in V_{max} and k. The highest values of V_{max} (0.73 g h^{-1} L^{-1}) and k (0.71 h^{-1}) were obtained at an enzymatic concentration of 30X. However, as discussed previously, this increase in enzymatic concentrations did not entail significant increases in TC, GLU, or XYL yields, and therefore, the higher costs are not justified.

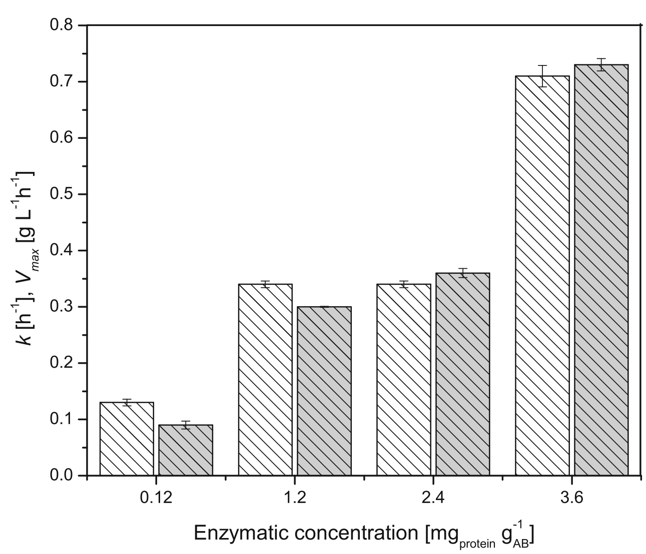


Fig. 3 Kinetic parameters of the test fitted by the Gompertz model: k (open), V_{max} (gray fill)

Therefore, we decided to continue the EH experiments with an enzymatic concentration of 2.4 $mg_{protein}$ $g^{-1}{}_{AB}$ (20X).

Global TC, GLU, and XYL Yields for Different Systems Tested

The global yields, i.e., those obtained after pretreatment and EH, are shown in Fig. 4. The contribution of pretreatment to the global TC yield ranged from 13 to 34% (Fig. 4a) for the different systems, while this contribution was marginal for GLU and XYL global yields (Fig. 4b, c). The highest global TC yield (0.7 g $g^{-1}{}_{AB}$) was obtained at SF 2.54 and was statistically similar to the global TC yield obtained at 2.41 and significantly higher than the global TC yield of the other systems. This result indicated that the pretreatment time did not have a significant effect on the TC yield since both pretreatments were performed at 142 °C, and the increment in SF was attributable to the increment in time from 15 to 20 min. However, the highest GLU yield (0.12 g $g^{-1}{}_{AB}$) was obtained at SFs 2.41 and 2.89 and was significantly higher than the GLU yield obtained at SFs 1.65 and 2.54. The global XYL yields remained between 0.010 and 0.038 g $g^{-1}{}_{AB}$; with increasing SF values, the XYL yields also increased, except for

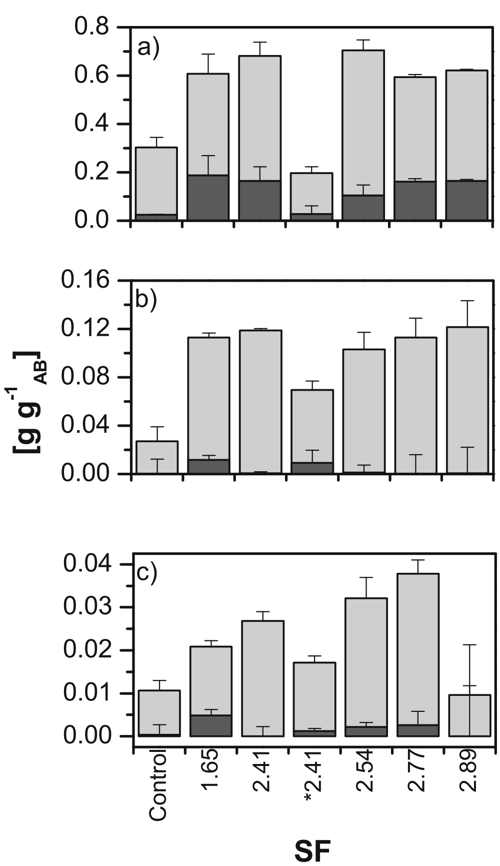


Fig. 4 Global yields of (**a**) TC, (**b**) GLU, and (**c**) XYL from pretreatment (dark gray) and enzymatic hydrolysis (light gray) as a function of SF. Enzyme concentration in all cases 2.4 $mg_{protein}$ $g^{-1}{}_{AB}$ of Cellic CTec2, except (*) 23.1 $mg_{protein}$ g^{-1} of Celluclast 1.5 L-Viscozyme L

the most severe conditions, which is attributable to the degradation of sugars due to higher temperatures and longer pretreatment times (154 °C and 20 min).

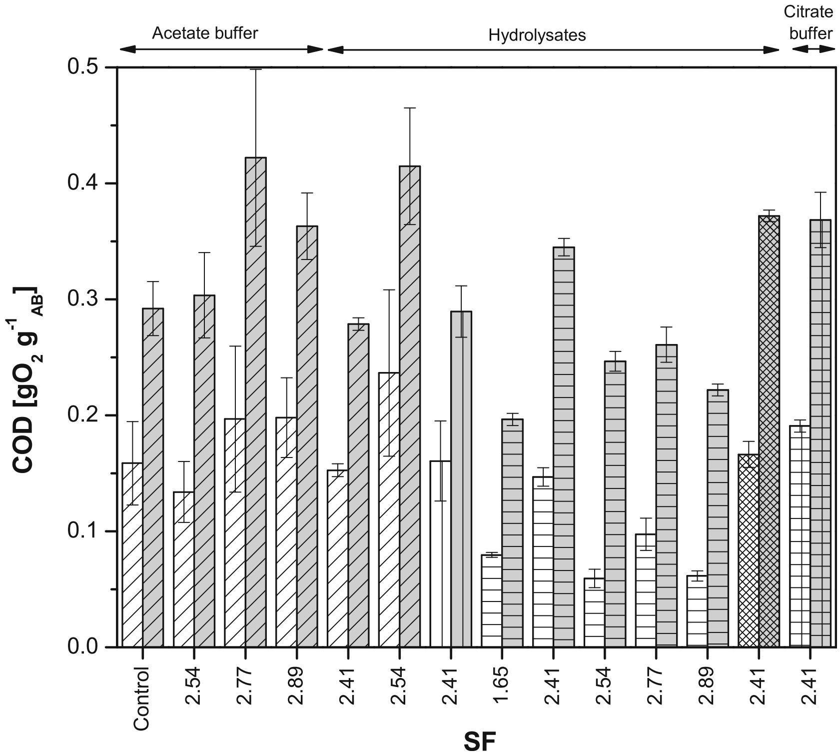
Therefore, the best yields were obtained with pretreatment at SF 2.41 and EH with Cellic CTec2 (2.4 $mg_{protein}$ $g^{-1}{}_{AB}$), and this system was compared to the system pretreated at the same SF and enzymatically hydrolyzed with Celluclast 1.5 L-Viscozyme L mixture (23.1 $mg_{protein}$ $g^{-1}{}_{AB}$) using previously optimized conditions. The analysis of Gompertz model parameters obtained for these two systems indicated that the values of V_{max} (g L^{-1} h^{-1}) and k (h^{-1}) for Celluclast 1.5 L-Viscozyme L mixture was 45% and 15% higher than the Cellic CTec2 system. In contrast, the TC, GLU, and XYL yields (g $g^{-1}{}_{AB}$) observed were significantly lower (38, 59, and 64%) than the TC, GLU, and XYL yields obtained with Cellic CTec2 (Fig. 4a, b, and c) despite the 9.6 times higher enzymatic concentration, which can be explained by the 2.6 lower ratio of AB to aqueous phase used in the Cellic CTec2 system.

In a previous study, 312.54 ± 46.89 mg of reducing sugars $g^{-1}{}_{AB}$ under the optimized conditions was reported [15]. However, in this study, only GLU and XYL were considered. The global TC yield obtained was 43 and 59% higher than the global TC yield reported for AB pretreated with alkaline hydrogen peroxide and sequential EH with Viscozyme L and Celluclast 1.5 L and AB only with binary EH [12, 25]. This value was also 49% higher than the value obtained by EH with Stonezyme [25]. In contrast, the highest global GLU yields were 57, 58, and 50% lower than the global GLU yields obtained with pretreated AB with ammonia fiber expansion, autohydrolysis, or ionic liquid and EH with Cellic CTec2 and Cellic HTec2, respectively [26]. The highest global yield of XYL was also significantly lower than the global yield of XYL obtained from ammonia fiber expansion, autohydrolysis, or ionic liquid pretreatment followed by EH with Cellic CTec2 and Cellic HTec2 [26]. TC yields higher than the yields previously reported but the GLU and XLY yields lower indicates that other carbohydrates were released. In the following section, the effects of these other carbohydrates on anaerobic digestion are discussed.

COD Evaluation

All cases exhibited an increase in COD with respect to the initial value (Fig. 5), indicating the release of organic matter to the aqueous phase. The pretreatments SFs 2.77 and 2.89 resulted in increases in the final COD of 44.56 and 24.35%, respectively, in relation to the control. However, the EH of the solid fraction from pretreatment SF 2.54 with 0.12 $mg_{protein}$ $g^{-1}{}_{AB}$ Cellic CTec2 showed no significant differences when compared to the control (without pretreatment). However, the final COD after EH with 0.12 $mg_{protein}$ $g^{-1}{}_{AB}$ of Cellic CTec2 in both fractions of the pretreatment at SF 2.54 was 27%

Fig. 5 Initial COD (open) and final COD (gray fill) yields. Enzymatic concentrations in mg$_{protein}$ g^{-1}$_{AB}$ of Cellic CTec2: 0.12 (▨), 1.2 (▥), 2.4 (▤), 3.6 (▩), and Celluclast 1.5 L-Viscozyme: 23.1 (▦)



higher than the EH of only the solid fraction. In contrast, the increase in enzymatic concentration up to 3.6 mg$_{protein}$ g^{-1}$_{AB}$ of Cellic CTec2 in systems with solid and liquid fractions pretreated at SF 2.41 did not entail an increase in the final COD, which may be related to the yields of GLU and XYL also not showing a significant increase from 2.4 to 3.6 mg$_{protein}$ g^{-1}$_{AB}$.

Regarding systems using both fractions after pretreatments at a fixed concentration of 2.4 mg$_{protein}$ g^{-1}$_{AB}$ of Cellic CTec2 and different SF values, the highest concentration of the final COD in the pretreatment SF 2.41 (0.34 ± 0.03 g O_2 g^{-1}$_{AB}$) was 41% higher than the highest concentration of the final COD in the pretreatment SF 1.65. The value is comparable to the value obtained using alkaline peroxide hydrogen as pretreatment and EH with Viscozyme L (22 mg$_{protein}$ g^{-1}$_{AB}$), as well as EH with Cellulase 50XL (~ 14 FPU g^{-1}$_{AB}$) and AB pretreated with rumen fluid or diluted acid [11, 12, 27]. In contrast, the value is lower than the 0.7 g O_2 g^{-1}$_{AB}$ obtained with EH (Celluclast 1.5 L, 0.7 mg$_{protein}$ g^{-1}$_{AB}$) and 1.1 g O_2 g^{-1}$_{AB}$ obtained after alkaline peroxide hydrogen and EH with Celluclast 1.5 L and Viscozyme L (14 and 2 mg$_{protein}$ g^{-1}$_{AB}$) [12, 23]. Nevertheless, no significant differences were seen between SF 1.65 and pretreatments having higher SF values (2.54, 2.77, and 2.89). The comparison of the COD in the systems hydrolyzed with Cellic CTec2 or Celluclast 1.5 L-Viscozyme L mixture showed no significant differences. The system with 2.4 mg$_{protein}$ g^{-1}$_{AB}$ and pretreatment SF 2.41 exhibited an increase in COD of 0.20 ± 0.01 g O_2 g^{-1}$_{AB}$ without significant differences with respect to higher enzymatic concentrations. Similar to the behavior for global TC, GLU, and XYL yields for different systems tested discussed above, a decrease in the COD obtained was observed with increasing SF.

BMP studies

Based on the results obtained, AB pretreated with SF 2.41 (142 °C for 15 min) was evaluated for BMP, and the results with and without EH are shown in Table 4, where these results are compared with data reported in the literature. The BMP of enzymatic hydrolysates of the pretreated materials was 68% greater than the BMP of the system without EH. In comparison, the BMP of the hydrolysate system with Cellic CTec2 was 17% greater than the BMP of the Celluclast 1.5 L-Viscozyme L mixture, and no significant differences were found ($p < 0.05$). The fit of the BMP tests by the Gompertz model is presented in Figure 1S in the Electronic Supplementary Material, while the kinetic parameters are reported in Table 5. The V_{max} obtained with the enzymatic hydrolysates using Cellic CTec2 was 45 and 20% higher than the V_{max} without HE and with the mixture Celluclast 1.5 L-Viscozyme L, respectively. However, t_r also increased, meaning that the maximum methane production requires a longer process time to achieve the maximum concentration (30 to 37% higher). During anaerobic digestion of enzymatic hydrolysates with Cellic CTec2, 78.4% removal of COD was achieved, slightly superior to the values previously reported (72.8 and 60%) [28, 29]. The value of BMP obtained was

Table 4 Comparison of BMP values reported in this study and in the literature for bagasse agave with different pretreatments and enzymes

Pretreatment type/conditions	Enzymatic hydrolysis enzyme/conditions	BMP ($NLCH_4$ g^{-1} COD_{add})	Reference
Steam explosion 142 °C, 15 min	—	0.169 ± 0.03	This study
Steam explosion 142 °C, 15 min	Cellic CTec2 2.4 $mg_{protein}$ $g^{-1}{}_{AB}$, 50 °C, 25 g_{AB} L^{-1}	0.284 ± 0.02	This study
Steam explosion 142 °C, 15 min	Celluclast 1.5 L-Viscozyme L 3.7/19.4 $mg_{protein}$ $g^{-1}{}_{AB}$, 40 °C, 64.84 g_{AB} L^{-1}	0.236 ± 0.05	This study
Steam explosion 142 °C, 5 cycles	—	0.260	[31]
Steam explosion 142 °C, 3 cycles	—	0.317	[31]
Steam explosion 158 °C, 20 min	—	0.168 ± 0.007	[11]
Steam explosion 178 °C, 24 min, 5 cycles	—	0.220	[11]
Acid hydrolysis 1.4% (wt/v) HCl, 123 °C, 126 min	—	0.260	[28]
Acid hydrolysis 2.7% (wt/v) HCl, 124 °C, 78 min	Celluclast 1.5 L 40 FPU $g^{-1}{}_{AB}$, 40 °C, 40 g_{AB} L^{-1}	0.240	[29]
Acid hydrolysis 1.8% (wt/v) HCl, 119 °C, 103 min	—	0.280	[30]
Acid hydrolysis 1.9% (wt/v) HCl, 130 °C, 193.2 min	—	0.230	[11]
Alkaline hydrogen peroxide 2% (wt/v), 50 °C, 90 min	Celluclast 1.5 L-Viscozyme 1.84 $mg_{protein}$ mL^{-1}, 0.1 $mg_{protein}$ mL^{-1}, 40 °C, 50 g_{AB} L^{-1}	0.200 ± 0.02	[12]
Biological pretreatment with rumen fluid 25 g_{AB} L^{-1} S_0/X_0 ratio 0.33 (g_{vs}/g_{vs}) 37 °C, 15 days	—	0.175	[11]
Biological pretreatment with native microbiota 40.5 g_{AB} L^{-1} S_0/X_0 ratio 0.7 (g_{vs}/g_{vs}) 37 °C, 4 days	—	0.241	[11]
—	Cellulase 50 XL 10.67 $mg_{protein}$ $g^{-1}{}_{AB}$, 42.7 °C, 87.5 g_{AB} L^{-1}	0.220	[11]
Ozone pretreatment 90 mg O_3 $g^{-1}{}_{AB}$, 60 min	Celluclast 1.5 L 4.72 UI $g^{-1}{}_{AB}$, 50 °C, 50 g_{AB} L^{-1}	0.210	[11]
Liquid ionic [Ch][Lys] 20% (wt/v) 124 °C, 205 min	Cellic CTec2 8 FPU $g^{-1}{}_{AB}$, 50 °C, 40 g_{AB} L^{-1}	0.300	[32]

N, normalized data (1 atm and 0 °C); *BMP*, biochemical methane potential; *FPU*, filter paper units; *S0/X0*, initial substrate/biomass; *[Ch][Lys]*, cholinium lysinate; —, not applicable; *vs*, volatile solids

0.284 NL CH_4 g^{-1} COD, comparable to the values found in other studies where acid hydrolysis was used as pretreatment [28–30]. In this sense, the pretreatment used herein has the advantage of not employing acid catalysts and of using a ratio of 25 g_{AB} per liter, while in the studies mentioned above, ratios between 40 and 50 g_{AB} per liter were used. However, the BMP values were 10% lower than the values obtained when three cycles of steam explosion were used and 6% lower than the values reported for pretreatment with ionic liquids [31, 32]. However, compared to alkaline and biological pretreatments, the BMP obtained herein was between 15 and 38% greater [11, 12].

The steam explosion conditions employed in this study, compared with the conditions in the literature for AB pretreatment, have the advantage of a shorter pretreatment time and, in some cases, a lower temperature without the use of catalysts, possibly resulting in diminished environmental impacts, but the achievement of similar BMP values requires the application of EH. Since the BMP values obtained with different pretreatments with and without EH, in general, reached the same range, the differentiation between them should include data on operating time and costs, the energy use required, and its performance in anaerobic digestion reactors.

Conclusions

In all cases, pretreatment resulted in higher TC yields after EH, indicating the relevance of this first step for the use of AB for biofuel production. The systems where both fractions generated by the pretreatments were used for EH favored higher TC and XYL yields. The pretreatment conditions that favored the highest TC, GLU, and XYL yields after EH did not correspond to the most severe conditions, implicating a reduction in the energy and time invested in this process and directly affecting the costs of biogas production. Likewise, the increase in enzymatic concentration of CellicTec2 enhanced the yields of TC, GLU, and XYL, as well as the V_{max}. However, there were no significant differences between the two highest enzymatic concentrations tested. Furthermore, pretreatment and EH significantly increased the BMP values of the hydrolysates compared to the BMP values of the controls (without EH or pretreatment). The BMP values obtained for the hydrolysates are comparable to the values reported in the literature for other pretreatment systems such as acid hydrolysis, biological and chemical systems, and other enzymatic formulations.

Table 5 Kinetic parameters of the BMP test fitted by the Gompertz model

Enzymatic cocktail	R^2	k (h^{-1})	V_{max} (NmL h^{-1})	t_r (h)
None (control)	0.984	0.048 ± 0.001	7.4 ± 4x10^{-4}	16.3 ± 0.3
Cellic Ctec 2	0.986	0.056 ± 0.001	13.4 ± 5x10^{-4}	21.3 ± 0.3
Celluclast 1.5 L-Viscozyme L	0.997	0.063 ± 0.001	10.8 ± 3x10^{-4}	19.7 ± 0.1

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Author Contributions V Duran-Cruz carried out the research, analyzed the data, prepared the visuals, and wrote the first draft.

S Hernández supervised the pretreatment experimentation and analyzed the data.

I Ortíz did the conceptualization, supervised the research, analyzed the data, and wrote the final draft.

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Compliance with Ethical Standards

Conflict of Interest The authors declare that they have no conflict of interest.

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